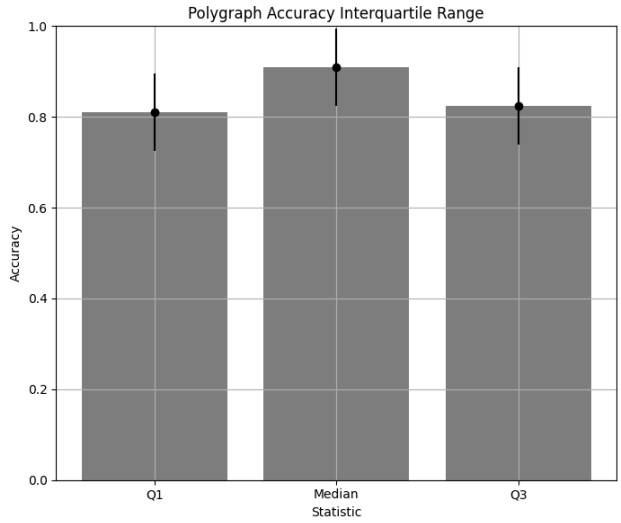
Site: https://www.kaggle.com/datasets/washingtonpost/police-shootings/data
https://www.kaggle.com/code/hasibalmuzdadid/killings-by-law-enforcement-officer-analysis

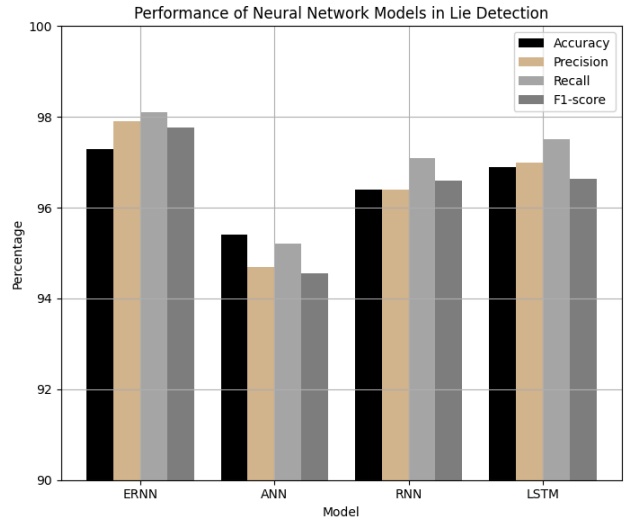
Datasets for Deception / Lie Detection
https://www.kaggle.com/datasets/thesergiu/truth-detectiondeception-detectionlie-detection
https://onlinelibrary.wiley.com/doi/full/10.1002/acp.3777

ChatGPT Datasets
https://nap.nationalacademies.org/read/10420/chapter/7
https://www.mdpi.com/2078-2489/14/10/564
https://link.springer.com/article/10.1007/s11042-023-16769-w

Criminal DataSets:
https://liedetectortest.com/police-polygraph/lie-detector-tests-that-have-resolved-crimes#:~:text=Dill%20had%20been%20a%20suspect,than%20he%20had%20previously%20admitted.

Polygraph Test:
https://pubmed.ncbi.nlm.nih.gov/31282623/





Polygraph Accuracy from the National Academies Press:

- The bar graph on the left shows the interquartile range (Q1 to Q3) of polygraph accuracy values with an added median value. The error bars represent the standard deviation of these accuracy values, indicating variability in the data. The

interquartile range (Q1: 0.81, Q3: 0.91) highlights that most polygraph tests have accuracy values within this range.

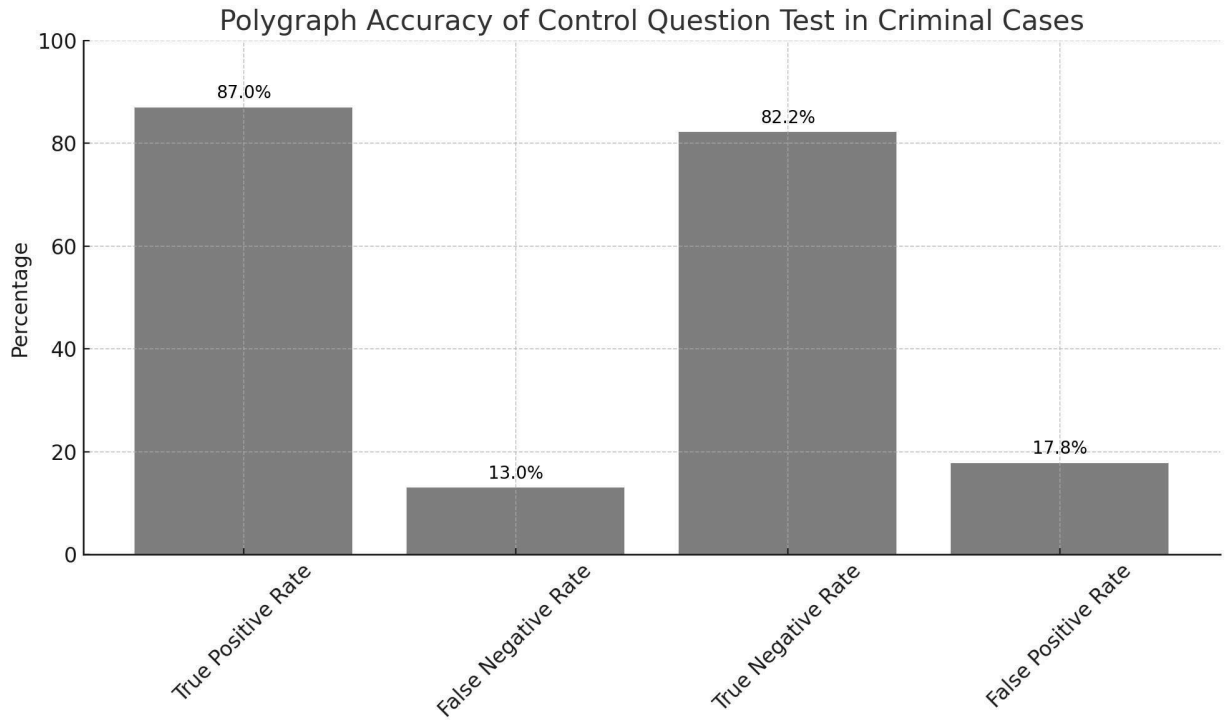
Performance of Neural Network Models from Springer:

- The bar graph on the right compares the performance of different neural network models (ERNN, ANN, RNN, LSTM) used in lie detection through voice stress analysis. The metrics displayed include accuracy, precision, recall, and F1-score. The ERNN model consistently outperforms the others across all metrics, demonstrating its superior capability in detecting deception.

he abbreviations ERNN, ANN, RNN, and LSTM stand for different types of neural network models:

1. **ERNN**: Explainable Enhanced Recurrent Neural Network
    - A variant of RNN that focuses on providing explainability alongside improved performance.
2. **ANN**: Artificial Neural Network
    - A computational model inspired by the way biological neural networks in the human brain process information. It consists of layers of interconnected nodes (neurons) that process data in a hierarchical manner.
3. **RNN**: Recurrent Neural Network
    - A type of neural network where connections between nodes can create cycles, allowing the network to maintain a form of memory. This makes RNNs particularly well-suited for sequential data, such as time series or text.
4. **LSTM**: Long Short-Term Memory
    - A special kind of RNN designed to better handle long-term dependencies. LSTMs use memory cells that can maintain information in memory for long periods, making them effective for tasks requiring the understanding of context over time.

For the Polygraph Test:



About Statement:

Our mission is to rigorously investigate and enhance the effectiveness of lie detection methods in criminal cases. We are dedicated to ensuring the integrity and reliability of truth verification techniques, aiming to bolster the judicial process and uphold justice.

By conducting comprehensive research, analyzing data, and developing advanced methodologies, we strive to improve the accuracy and credibility of lie detection tools such as polygraph tests. Our commitment to scientific excellence and ethical standards drives us to provide law enforcement, legal professionals, and the broader community with reliable information and innovative solutions for detecting deception.

We believe that through meticulous research and continuous improvement, we can contribute to a more just and transparent legal system, ensuring that truth prevails in the pursuit of justice.

Citations

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